PYTHON 90 DAYS CHALLENGE

DAY 09 — break & continue

GitHub-ready notes • Coding practice • Interview questions

Based on the uploaded Python notes: Chapter 8, pages 111–112 (break and continue).

Learning goals

• Understand how **break** exits a loop.
• Understand how **continue** skips the current iteration.
• Practise tracing loop output and writing small programs.
• Explain both statements in a beginner interview.

1. The break Statement

Meaning: `break` immediately stops the nearest loop. The program then continues with the statement after that loop.

Example: Stop when the number reaches 4

```
for number in range(1, 8):
    if number == 4:
        break
    print(number)
```

Output

```
1
2
3
```

Why? When `number` becomes 4, `break` exits the loop before 4 is printed.

2. The continue Statement

Meaning: `continue` skips the rest of the current iteration and moves to the next iteration. It does not end the whole loop.

Example: Skip number 3

```
for number in range(1, 6):
    if number == 3:
        continue
    print(number)
```

Output

```
1
2
4
5
```

Why? When `number` is 3, `continue` skips `print(number)` for that iteration. The loop continues with 4 and 5.

3. break vs continue — Quick Revision

Point	break	continue
Effect	Stops the loop	Skips current iteration
Loop continues?	No	Yes, with next iteration
Typical use	Stop when target is found	Skip unwanted values
Example result	1, 2, 3 (stop at 4)	1, 2, 4, 5 (skip 3)

4. Coding Practice — Try Before Checking

For every question, first write the code yourself. The expected output is included so you can test your solution.

Q1. Print numbers from 1 to 10, but stop before printing 6.

Expected output: 1 2 3 4 5

Q2. Print numbers from 1 to 10, skipping all numbers divisible by 3.

Expected output: 1 2 4 5 7 8 10

Q3. Take an integer n and stop the loop when i becomes n. Print numbers before n.

Example: n = 5 → output: 1 2 3 4

Q4. Print only odd numbers from 1 to 10 by using continue to skip even numbers.

Expected output: 1 3 5 7 9

Q5. Search for the first number from 1 to 20 that is divisible by both 4 and 6. Print it and stop.

Expected output: 12

Q6. Print each character in the word 'PYTHON', but skip the letter 'H'.

Expected output: P Y T O N

Q7. Given numbers = [4, 7, -2, 9, 0, 11], print values until the first negative number appears.

Expected output: 4 7

Q8. Print numbers from 1 to 15, skipping multiples of 4 and stopping after printing 11.

Expected output: 1 2 3 5 6 7 9 10 11

5. Sample Solutions

Check these only after attempting the questions.

Q1

```
for i in range(1, 11):
    if i == 6:
        break
    print(i, end=" ")
```

Q2

```
for i in range(1, 11):
    if i % 3 == 0:
        continue
    print(i, end=" ")
```

Q3

```
n = int(input("Enter n: "))
for i in range(1, n + 1):
    if i == n:
        break
    print(i, end=" ")
```

Q4

```
for i in range(1, 11):
    if i % 2 == 0:
        continue
    print(i, end=" ")
```

Q5

```
for i in range(1, 21):
    if i % 4 == 0 and i % 6 == 0:
        print(i)
        break
```

Q6

```
for ch in "PYTHON":
    if ch == "H":
        continue
    print(ch, end=" ")
```

Q7

```
numbers = [4, 7, -2, 9, 0, 11]
for value in numbers:
    if value < 0:
        break
    print(value, end=" ")
```

Q8

```
for i in range(1, 16):
    if i % 4 == 0:
        continue
    print(i, end=" ")
    if i == 11:
        break
```

6. Python Interview Questions & Answers

1. What is break in Python?

Answer: break is a loop-control statement that immediately terminates the nearest enclosing loop.

2. What is continue in Python?

Answer: continue skips the remaining statements in the current loop iteration and proceeds to the next iteration.

3. What is the main difference between break and continue?

Answer: break ends the loop; continue skips only the current iteration.

4. Can break be used inside a for loop and a while loop?

Answer: Yes. It can be used in both for and while loops.

5. Does continue terminate a loop?

Answer: No. It skips the current iteration, and the loop can continue with the next one.

6. What happens to code written after a loop containing break?

Answer: If execution reaches the break, the loop ends and execution continues with the statement after the loop.

7. What does this code print?

```
for i in range(1, 5):
if i == 3:
break
print(i)
```

Answer: It prints 1 and 2.

8. What does this code print?

```
for i in range(1, 5):
if i == 3:
continue
print(i)
```

Answer: It prints 1, 2, and 4.

9. Can break be used outside a loop?

Answer: No. Using break outside a loop causes a SyntaxError.

10. Give a practical use case for break.

Answer: Stop searching a list when the required item is found, or stop input processing when a sentinel value is entered.

7. GitHub README Template

Copy this into a README.md file in your Day 09 folder:

```
# Python 90 Days Challenge
## Day 09 - break and continue

### Topics Covered
- break statement
- continue statement
- Difference between break and continue
- Loop output tracing

### Practice
Completed beginner exercises using for loops, conditions, break, and continue.

### How to Run
1. Install Python 3.
2. Save the code in `day09_break_continue.py`.
3. Run: `python day09_break_continue.py`

### Key Takeaway
```

```
- `break` exits the loop.
- `continue` skips the current iteration.
```

Suggested GitHub file names

```
day-09-break-continue.pdf
day09_break_continue.py
README.md
```

Revision checklist

- ■ I can explain break in my own words.
- ■ I can explain continue in my own words.
- ■ I can predict the output of both examples.
- ■ I attempted all 8 coding questions.
- ■ I practised the interview answers aloud.

Source alignment: the notes in this PDF follow the uploaded file's explanation and examples for break (page 111) and continue (page 112). Extra practice and interview questions are added for learning.